

$$\begin{array}{ccccc}
 X \iff B \wedge C & & X \implies B \wedge C & & \neg X \vee (B \wedge C) \text{ --- } \left[\begin{array}{l} \overset{2}{\neg X \vee B} \\ \overset{3}{\neg X \vee C} \end{array} \right.
 \end{array}$$

$$\begin{array}{cccc}
 \overset{1}{A \vee C} & & B \wedge C \implies X & & \neg(B \wedge C) \vee X \qquad \qquad \qquad \overset{4}{\neg B \vee \neg C \vee X}
 \end{array}$$

$$S = \{ \overset{1}{}, \overset{2}{}, \overset{3}{}, \overset{4}{} \}$$

$$\begin{array}{l}
 v(A) = true \\
 v(B) = false \\
 v(C) = false \\
 v(X) = true
 \end{array}
 \implies
 \left\{ \begin{array}{l} v \models F \\ v \not\models S \end{array} \right.$$

$$\begin{array}{l}
 v(A) = true \\
 v(B) = false \\
 v(C) = false \\
 v(X) = false
 \end{array}
 \implies
 \left\{ \begin{array}{l} v \models F \\ v \models S \end{array} \right.$$